

Question			Marking details	Marks available																				
				AO1	AO2	AO3	Total	Maths	Prac															
5	(a)		For a system to be in equilibrium (1) $\sum$ anticlockwise moments [about a point] = $\sum$ clockwise moments [about the same point] (1) N.B. Award 1 mark for $\sum$ C.M = $\sum$ A.C.M. only Alternative: For a system to be in equilibrium (1) algebraic sum of moments / net moment / resultant moment [about a point] = 0 (1)	2			2																	
	(b)		Centre of gravity or line of action of weight remained inside lower surface of block [or equiv, e.g. CoG remains to left of pivot] Accept arrow clearly shown on diagram [1] Produces anticlockwise <u>moment</u> (or torque) [about pivot] [1]		2		2		2															
	(c)	(i)	All mean F values determined correctly [1] <table><tr><td>Mean F / N</td><td rowspan="7">Or accept:</td><td>Mean F / N</td></tr><tr><td>2.7</td><td>2.7[0]</td></tr><tr><td>5.5</td><td>5.45</td></tr><tr><td>8.7</td><td>8.65</td></tr><tr><td>11.4</td><td>11.4[0]</td></tr><tr><td>14.6</td><td>14.55</td></tr><tr><td>17.4</td><td>17.4[0]</td></tr></table>	Mean F / N	Or accept:	Mean F / N	2.7	2.7[0]	5.5	5.45	8.7	8.65	11.4	11.4[0]	14.6	14.55	17.4	17.4[0]		1		1	1	1
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		(ii)	$FL = W_B d + 490 W_R \text{ (1)}$ Manipulation and use of $L = 980$ to show $F = \frac{W_B d}{L} + \frac{490 W_R}{980} \text{ (1)}$ Alternative: $FL = W_B d + W_R \frac{L}{2} \text{ (1)}$ $\therefore \left[FL = W_B d + W_R \frac{L}{2} \right] \rightarrow F = \frac{W_B d}{L} + \frac{W_R}{2} \text{ (1)}$		2		2	2	2

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		(iii)	Titles and units on the axis correct i.e. (Mean) force or F / N, distance or d / mm (1) Suitable scales chosen so that the data points occupy at least $\frac{1}{2}$ of each axis and not involving awkward factors, e.g. 3 (1) allow ecf from table All points plotted correctly to within $\pm \frac{1}{2}$ small square division ecf from table (1) Line of best fit drawn correctly not through origin (1) mean F/N <table><caption>Data points from the graph</caption><tr><th>distance d/mm</th><th>mean F/N</th></tr><tr><td>100</td><td>2.5</td></tr><tr><td>250</td><td>5.5</td></tr><tr><td>400</td><td>8.5</td></tr><tr><td>550</td><td>11.5</td></tr><tr><td>700</td><td>14.5</td></tr><tr><td>850</td><td>17.5</td></tr></table> distance d/mm	distance d/mm	mean F/N	100	2.5	250	5.5	400	8.5	550	11.5	700	14.5	850	17.5				4		4	4	4
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		(iv)	I.	Gradient calculated correctly e.g. $\frac{12.4 - 0.6}{600} = 0.02$ [1] $W_B = \text{gradient ecf} \times 980$ e.g. = approx 19.6 [N] [1]			2	2	2	2
			II.	y-intercept determined correctly $\pm$ small square tolerance e.g. 0.7 [N] [1] $W_R = 2 \times y\text{-intercept}$ e.g. = 1.4 N ecf but don't apply for intercept of 0 [1]			2	2	2	2
	(d)			% Unc (d) calculated i.e. $1 \times \frac{100}{100} = 1\%$ [1] Uncertainty in (spread) $\Delta F = \frac{2.8 - 2.6}{2} = 0.1$ [1] $p_F = \frac{0.1 \text{ecf}}{2.7} \times 100\% = 3.7\%$ [1] Ignore s.f.	1	1 1		3	3	3
	(e)			Any $1 \times (1)$ of [Spirit] level / appropriate method to ensure that the ruler is horizontal Use digital forcemeter / forcemeter with a higher resolution Clamp stand to hold newtonmeter Repeat readings <u>of force</u> [due to random errors]			1	1		1
				Question 5 total	3	11	5	19	14	17